

Qingxiao Huang

PhD Student · Department of Computer Science
City University of Hong Kong

qx.huang@my.cityu.edu.hk · [qxhuang0322.github.io](https://github.com/qxhuang0322)
[Google Scholar](#) · [ORCID 0009-0005-3883-5112](#) · [CityU Scholars](#)

Education

City University of Hong Kong PhD in Computer Science Supervisor: Prof. Yuguang "Michael" Fang	2025 - Present
University of Electronic Science and Technology of China Master's in Information and Communication Engineering Supervisor: Prof. Jie Hu · Collaborator: Prof. Kun Yang	2022 - 2025
Xidian University Bachelor's in Communication Engineering	2018 - 2022

Research Interests

Wireless Communications · Reconfigurable Intelligent Surfaces · Holographic MIMO · Near-Field Communications

Intelligent Surfaces. Reconfigurable and specular reflecting surfaces for shaping wireless propagation, with beamforming and multicast transmission in conventional and terahertz bands.

Holographic & Near-Field MIMO. Electromagnetic modeling and beamforming for continuous and circular holographic arrays, including spatial orthogonality and near-field energy focusing.

Wireless Information & Power. Integrated data and energy transfer through joint beamforming design, balancing communication performance and wireless energy-harvesting requirements.

Selected Honors & Awards

Research Tuition Scholarship · City University of Hong Kong	2026
Outstanding Graduate of Sichuan Province · Sichuan Province · Class of 2025	2025
National Scholarship for Graduate Students · Ministry of Education of China	2024
Outstanding Master's Thesis Award · University of Electronic Science and Technology of China	
Outstanding Graduate of UESTC · University of Electronic Science and Technology of China	2024
Academic Star Award · School of Information and Communication Engineering, UESTC	2024
Young Scholar Award · University of Electronic Science and Technology of China	2024

Academic Service & Teaching

Technical Program Committee. IEEE VTC2025-Fall Workshops · TPC Member

Teaching Assistant. Computer Science, City University of Hong Kong; Semester A, 2026-2027

Publications & Preprints

6 journal articles · 2 preprints · Name in bold identifies Qingxiao Huang.

Journal Articles

J1. CA3D: Computing Accessibility-Aware Cooperative 3D Deployment of Multiple UAVs

Yiqin Deng, Zihan Fang, Yijie Wang, **Qingxiao Huang**, Junhui Gao, Qian Yao Ren, Yuguang Fang

IEEE Wireless Communications Letters, 2026.

DOI: [10.1109/LWC.2026.3723451](https://doi.org/10.1109/LWC.2026.3723451)

J2. Circular Holographic MIMO Beamforming for Integrated Data and Energy Multicast Systems

Qingxiao Huang, Yizhe Zhao, Jie Hu, Kun Yang, Yuguang Fang

IEEE Transactions on Wireless Communications, 2026.

DOI: [10.1109/TWC.2025.3620886](https://doi.org/10.1109/TWC.2025.3620886)

J3. Intelligent Reflecting Surface Assisted Integrated Data and Energy Multicast System in Terahertz-Bands

Qingxiao Huang, Jie Hu, Kun Yang

IEEE Transactions on Green Communications and Networking, 2025.

DOI: [10.1109/TGCN.2024.3415030](https://doi.org/10.1109/TGCN.2024.3415030)

J4. Holographic Integrated Data and Energy Transfer

Qingxiao Huang, Jie Hu, Yizhe Zhao, Kun Yang

IEEE Transactions on Wireless Communications, 2024.

DOI: [10.1109/TWC.2024.3477750](https://doi.org/10.1109/TWC.2024.3477750)

J5. Beamforming Design in Intelligent Reflecting Surface Aided Multicast System in Ultra-High Frequency Bands With Large-Scale Antenna Array

Qingxiao Huang, Jie Hu, Qingdong Yue, Kun Yang

IEEE Wireless Communications Letters, 2024.

DOI: [10.1109/LWC.2024.3373494](https://doi.org/10.1109/LWC.2024.3373494)

J6. Hybrid Terahertz Beamforming Relying on Channel Sparsity and Angular Orthogonality

Qingdong Yue, Jie Hu, Tingyu Shui, **Qingxiao Huang**, Kun Yang, Lajos Hanzo

IEEE Transactions on Vehicular Technology, 2024.

DOI: [10.1109/TVT.2023.3332206](https://doi.org/10.1109/TVT.2023.3332206)

Preprints

P1. Bringing Environmental Enhancement Back to Its Physical Essence via Specular Reflecting Surfaces

Qingxiao Huang, Qian Yao Ren, Yiqin Deng, Jun Huang, Kun Yang, Yuguang Fang

arXiv preprint arXiv:2608.15271, 2026.

DOI: [10.48550/arXiv.2608.15271](https://doi.org/10.48550/arXiv.2608.15271)

P2. Shatter Throughput Ceilings: Leveraging Reflection Surfaces to Enhance Transmissions for Vehicular Fast Data Exchange

Qian Yao Ren, **Qingxiao Huang**, Yiqin Deng, Xianhao Chen, Phone Lin, Yuguang Fang

arXiv preprint arXiv:2603.02752, 2026.

DOI: [10.48550/arXiv.2603.02752](https://doi.org/10.48550/arXiv.2603.02752)